RANDOM NUMBERS



LECTURE OUTLINE

Generating Random Numbers

Class Activities for Random numbers

GENERATING RANDOM NUMBERS

- *You can use the* **rand()** *function to obtain a random integer.*
- This function returns a random integer between 0 and RAND_MAX. RAND_MAX is platform-dependent constant.
- The rand() function's algorithm uses a value called the *seed* to control how to generate the numbers. By default, the seed value is 1.
- If you change the seed to a different value, the sequence of random numbers will be different. To change the seed, use the srand(seed) function in the cstdlib header file.
- srand(time(0)); produces a different output every time you run the program.

GENERATING RANDOM NUMBERS:

- To obtain a random integer between 0 and 9, use rand() % 10
- To obtain a random integer between 0 and 99, use rand() % 100

ACTIVITY: GENERATING RANDOM NUMBERS

- The program may be set up to work as follows:

  Step 1: Generate two single-digit integers into number1 and number2.

  Step 2: Ask user to enter y if he/she wants to play this game.

  Step 3: if user enters yes Prompt the user to answer “What is number1 + number2?”

  Step 4: Check the student’s answer and display whether it is correct. If user guess correct answer, display Congratulations otherwise print sorry

```
#include<iostream>
#include<cstdlib>
using namespace std;
main()
{    char ch;
     cout<<" would you like to play game press y to play \n";
     cin>>ch;
     if(ch=='y'||ch=='Y')
     {

     cout<<" we have generate two random numbers for you\n";
     int num1;
     int num2;
     int ans;
     num1=rand()%10;
     num2=rand()%10;
     cout<<" what would be result after adding these numbers\n";
     cout<<" Enter your answer\n\t";
     cin>>ans;
     int res=num1+num2;
          if(ans==res)
          {
          cout<<" congratulations :-)";
          }
          else
          {
          cout<<" sorry! Correct ans is "<<res<<endl;
          }
     }
     else
     {
     cout<<" You dont want to play\n Thanks for visiting us ";
     exit(0);
     }
}
```

GENERATING RANDOM NUMBER BETWEEN LOWER AND UPPER BOUND

Example:
Generating random numbers between 1 and 6 where 1 = minimum value and 6 = maximum value

Formula = (rand()%(maximum value-minimum value+1))+minimum value

TASK S

- Which of the following is a possible output from invoking rand()? 323.4, 5, 34, 1, 0.5, 0.234

a. How do you generate a random integer i such that 0<=*i* < 20?

b. How do you generate a random integer i such that 10<=*i*<=20?

c. How do you generate a random integer i such that 10 <=*i*<=50?

Solution for c

low=10

high =

50

rand()%

(high-

low+1)+l

TASK

- Write a program which ask user to generate a random number between 1 and 3 and print apple if random number is 1, prints, grapes if random number is 2 and prints peach if number is 3 (note use switch)

NEXT LECTURE

For loop

While loop